
Decodable but Directionally Misaligned: Auditing Distance-Based OOD Ranking on Domain-Adversarial Skin-Lesion Representations

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Abstract

Distance-based out-of-distribution (OOD) scoring assumes not only that a learned representation separates source and shifted inputs, but also that the induced ordering has the correct operational direction: larger distance should indicate greater OOD likelihood. We audit this assumption within a domain-adversarial representation-learning regime using an orthogonality-strength ladder. Three checkpoint families share the same architecture and 16-dimensional representation and differ only in $\lambda_{\text{orth}} = 0, 1, 5$. Covariance conditioning changed substantially across the ladder: the shared within-class covariance condition number increased by approximately two orders of magnitude (Kendall’s $\tau = 0.84$, exact $p = 2.8 \times 10^{-5}$), whereas the other four geometry summaries tested showed no significant trend. Under the pre-specified higher-distance-is-more-OOD convention, Mahalanobis AUROC (ISIC-test vs. PAD-UFES) remained near 0.40 at every ladder level, and cosine-to-centroid and pooled k -nearest-neighbor scorers showed the same pattern. Reversing score orientation corresponds to an orientation-free separability of only 0.58–0.60, showing that the distances retain weak domain-separation signal but systematically rank PAD-UFES samples as more source-typical than ISIC-test samples. In contrast, supervised probes used only as existence tests recovered domain membership from the identical embeddings at 0.72–0.81 AUROC. The supported finding is therefore a scoped dissociation between domain decodability and the weak, directionally misaligned ordering imposed by the tested source-fitted distance rules. Power and bootstrap analyses show that the $n = 13$ design can exclude only large monotonic associations; moderate effects remain unresolved. The finding neither shows that scalar distances contain no domain-separation signal nor establishes that domain-adversarial training or orthogonality regularization causes the misalignment.

1 Introduction

Deep neural networks are typically evaluated by predictive accuracy, yet deployment safety also depends on how a learned representation interfaces with downstream reliability mechanisms. A distance-based OOD score must satisfy two distinct requirements. First, it must retain some ability to separate source and shifted inputs. Second, that separation must have the correct operational orientation: under the usual distance convention, shifted inputs should receive larger scores. A score can therefore contain distribution-separation signal while remaining unusable as specified if it orders the two distributions in the wrong direction. What

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matters is not “geometry” in the undifferentiated sense, but the particular geometric structure and ordering to which a scoring functional is sensitive. Covariance conditioning, class separation, local neighborhoods, angular relationships, and distributional shape can move differently under training; a change in one of these properties need not imply a corresponding change in OOD score ordering.

Distance-based OOD estimators make this dependence explicit. Mahalanobis distance (Lee et al., 2018), cosine-to-centroid distance, and non-parametric k -nearest-neighbor scoring (Sun et al., 2022) all read a test input’s location relative to the training distribution, but they preserve different invariances and discard different aspects of representation structure. Moreover, AUROC depends only on score ordering: any strictly increasing transformation leaves AUROC unchanged, whereas reversing score orientation maps an AUROC a to $1 - a$. An AUROC below 0.5 is therefore not evidence that the scalar score contains no distribution-separation signal. It instead indicates that, under the stated positive-class and score-direction convention, the score ranks the two groups predominantly in the opposite direction. We consequently distinguish throughout between *directed OOD AUROC*, evaluated under the pre-specified higher-distance-is-more-OOD convention, and *orientation-free separability*, summarized descriptively as $\max(\text{AUROC}, 1 - \text{AUROC})$. The latter is a diagnostic of separation, not a post-hoc corrected detector: a polarity chosen after observing evaluation labels cannot be claimed as an operational calibration.

Domain-adversarial representation learning (Ganin & Lempitsky, 2015; Ganin et al., 2016) and orthogonality-based shared/private separation objectives (Bousmalis et al., 2016) provide a useful setting in which to audit this issue because they deliberately reorganize representation structure associated with nuisance variables such as acquisition site or imaging domain. We study an orthogonality-strength ladder within a domain-adversarial regime ($\lambda_{\text{orth}} = 0, 1, 5$), holding architecture and representation dimensionality fixed across its levels. The design supports statements about how measured quantities vary with orthogonality strength within this regime; because every checkpoint already belongs to the domain-adversarial regime and the available conventional baseline is not matched in architecture or dimensionality, it does not by itself identify either domain-adversarial training or orthogonality regularization as the cause of any observed OOD behavior.

Our audit separates four questions that are often conflated: (i) which measured aspects of representation structure vary across the orthogonality ladder; (ii) whether the directed OOD ranking produced by several distance-based scorers changes with those measured properties; (iii) whether the scalar scores retain any orientation-free domain separability; and (iv) whether domain membership remains recoverable from the full embeddings by a supervised reader. This separation prevents a below-chance directed AUROC from being misread as absence of information and prevents a label-informed score reversal from being presented as a deployable correction.

Across the ladder, covariance conditioning changes substantially while the other four geometry summaries tested show no significant trend. Five variants from three structurally distinct distance-based scorer families remain near 0.40–0.42 directed AUROC under the pre-specified OOD convention. Their corresponding orientation-free separability is only 0.58–0.60: the scores retain weak separation signal, but their ordering is systematically opposite to the intended operational direction. Supervised probes used as decodability existence tests recover substantially stronger domain membership information at 0.72–0.81 AUROC from the full embeddings. The central finding is therefore a scoped dissociation between domain decodability and the weak, directionally misaligned ordering delivered by the tested source-fitted distance rules within the audited domain-adversarial training regime, not a claim that distances contain no information and not a causal claim that disentanglement creates the misalignment.

2 Related Work

Post-hoc OOD detectors are usually grouped by whether they read the classifier’s output layer or its internal representation. Output-layer scores include maximum softmax probability (Hendrycks & Gimpel, 2017), ODIN (Liang et al., 2018), energy-based scoring (Liu et al., 2020), ReAct (Sun et al., 2021), ViM (Wang et al., 2022), and activation shaping (Djurisic et al., 2023); training-time objectives such as ELogitNorm (Ding et al., 2026) instead reshape representations so that such scores work better. Distance-based scorers read the penultimate representation directly: Mahalanobis distance (Lee et al., 2018), pooled k -nearest-

neighbor distance (Sun et al., 2022), nearest-neighbor guidance (Park et al., 2023), and Gram-matrix scoring (Sastry & Oore, 2020). Benchmarking across this family (Yang et al., 2024; Miyai et al., 2025) compares detectors across architectures and datasets under a fixed training regime. We hold architecture and dataset fixed and ask the complementary question: whether one detector’s behavior on one representation changes when that representation is deliberately reshaped by training.

Domain-adversarial training (Ganin & Lempitsky, 2015; Ganin et al., 2016), domain separation networks (Bousmalis et al., 2016), correlation alignment (Sun & Saenko, 2016), and adversarial discriminative adaptation (Tzeng et al., 2017) all suppress a nuisance variable in the representation; disentangled representation learning (Higgins et al., 2017; Locatello et al., 2019) pursues a related but distinct factorization. These methods are motivated by documented shortcut reliance in clinical imaging — scanner and hospital artefacts in chest radiographs (Zech et al., 2018; DeGrave et al., 2021), surgical skin markings in dermoscopy (Winkler et al., 2019), and above-chance accuracy on lesion-masked ISIC images (Bissoto et al., 2020) — synthesized by Geirhos et al. (2020). They are evaluated by downstream accuracy, or by a discriminator’s failure to recover the suppressed variable; to our knowledge, none is evaluated by whether a distance-based reliability estimator built on the resulting representation still behaves as it did before the intervention.

Probing measures what a frozen representation encodes (Alain & Bengio, 2016; Hewitt & Liang, 2019), with the standing caution that a successful probe does not show that the model itself uses the decoded information (Belinkov, 2022); we use probes only in that restricted, existence-test sense. A separate line characterizes representation geometry directly — intrinsic dimensionality (Levina & Bickel, 2004; Facco et al., 2017; Ansuini et al., 2019) and covariance conditioning — and Janiak et al. (2026) report that such summaries track Mahalanobis-style detector performance across frozen foundation-model backbones. That association is measured between representations differing simultaneously in pretraining corpus, objective, architecture, and scale, so it does not establish what happens when the same geometric quantities are moved within one fixed representation by a specific training intervention.

Finally, predictive uncertainty is known to degrade under dataset shift (Ovadia et al., 2019), with consequences for clinical deployment (Finlayson et al., 2021). Skin-lesion OOD detection has been studied with deep isolation-forest scoring (Li et al., 2020), benchmarked across covariate-shifted and far-OOD regimes (Gutbrod et al., 2025), and shown to leave classifiers overconfident (Pacheco et al., 2020b); skin tone is a distinct and largely orthogonal source of heterogeneity (Daneshjou et al., 2022) that our dataset pair does not separate from acquisition-device shift (Section 5.4). That literature evaluates reliability estimators against naturally occurring shift, not against a representation that is itself the target of an explicit debiasing intervention. Calibration (Guo et al., 2017), selective prediction (El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017), and conformal prediction (Angelopoulos & Bates, 2023) are outside the scope evaluated here. Appendix E gives the extended treatment of all six adjacent literatures.

3 Methods

3.1 Study design, data, and the orthogonality-strength ladder

For each of 13 checkpoints we extract z_{lesion} embeddings for the ISIC training set and for the ISIC-test/PAD-UFES evaluation split. From these embeddings we compute geometry metrics on the fitted Mahalanobis parameters (Section 3.2), score the evaluation split with three distance-based reliability estimators built on the identical embeddings (Section 3.3), and train three supervised probes on the same embeddings to quantify domain-information decodability independent of any distance-based scorer (Section 3.4). Association between orthogonality strength, the measured representation summaries, reliability estimation, and decodability is then tested across the ladder using exact-permutation statistics (Section 3.5). Figure 3 in Appendix A diagrams the pipeline.

The source domain is the ISIC 2019 training set (Codella et al., 2019; Tschandl et al., 2018; Combalia et al., 2019) — 25,331 images over the eight diagnostic classes MEL, NV, BCC, AK, BKL, DF, VASC and SCC — and the target domain is PAD-UFES-20 (Pacheco et al., 2020a) (2,298 images). We inherit the split and preprocessing pipeline of the classifier this study audits: ISIC is partitioned by a fixed seeded stratified split into 16,211 training, 4,053 validation and 5,067 test images, and PAD-UFES is withheld from label

Table 1: Orthogonality-strength ladder within the audited domain-adversarial regime.

Family	λ_{orth}	Mechanism	Seeds available
<code>runA_grl</code>	0	domain-adversarial component only	42, 52, 62, 72, 82 (5)
<code>runB_orth1</code>	1	+ orthogonality term	42, 52, 62, 72, 82 (5)
<code>runB</code>	5	+ stronger orthogonality	42, 52, 62 (3)

supervision during training. PAD-UFES is not an independent target domain. The audited architecture’s domain-adversarial branch explicitly optimizes the representation to reduce discriminability between ISIC and PAD-UFES, so this domain is not independent of the optimization process.

The three checkpoint families come from a domain-adversarial architecture with lesion and context branches, share identical architecture, training recipe, and a 16-dimensional lesion representation, and differ only in the strength λ_{orth} of an orthogonality penalty applied between the two branches (Table 1), giving 13 family-seed checkpoints in total. Every checkpoint is referenced by an explicit file path, recorded with each result, rather than resolved by automated directory search; the path used for each of the 13 checkpoints is reported with the released data. We additionally report a single conventional non-adversarial ResNet-50 checkpoint, `baseline_soft`, differing from the ladder in architecture, representation dimensionality (2048-d vs. 16-d), and training recipe, as a descriptive reference rather than a fourth ladder rung.

3.2 Representation geometry metrics

Three metrics were pre-specified to represent complementary aspects of representation geometry — covariance conditioning, class separation, and multivariate normality — rather than to maximize empirical correlation with downstream performance. The **condition number** is the ratio of largest to smallest eigenvalue of the regularized shared within-class covariance used to fit the Mahalanobis estimator. The **Fisher ratio** is reported both as the Hotelling–Lawley trace $\text{tr}(\Sigma^{-1}S_B)$ and as a decoupled scalar companion $\text{tr}(S_B)/\text{tr}(S_W)$, which involves no matrix inverse and is therefore not entangled with condition number’s own estimation noise. **Mardia’s multivariate kurtosis** (Mardia, 1970) quantifies departure from the class-conditional Gaussian shape the Mahalanobis estimator assumes; its null distribution is calibrated by parametric bootstrap (200 resamples per checkpoint), because the classical closed-form asymptotic null does not apply once means and covariance are estimated in-sample from the same residuals being tested. All are computed on the identical fitted Mahalanobis parameters used for scoring (Section 3.3), not from an independently re-estimated covariance, so that geometry and reliability estimation describe the same fitted object. Appendix A.1 gives the full definitions.

3.3 Distance-based reliability scorers

Each test embedding is scored by three structurally distinct distance-based rules, applied to identical embeddings and an identical ISIC-train / ISIC-test / PAD-UFES split, for every one of the 13 checkpoints:

- **Mahalanobis distance:** the minimum, over the 8 ISIC classes, of the squared Mahalanobis distance from the test embedding to each class mean, under a shared covariance fit on the ISIC training set with regularization $\varepsilon = 10^{-5}$.
- **Cosine-to-centroid:** the minimum, over the same 8 class means, of $1 - \cos(z, \mu_c)$ — identical class-centroid structure to the Mahalanobis scorer with the covariance/precision step removed, isolating that one variable.
- **Pooled k -nearest-neighbor distance:** the Euclidean distance from a test embedding to its k -th nearest neighbor in the full, class-pooled ISIC training set — no class structure, no covariance, the most assumption-free of the three (Sun et al., 2022). We treat $k = 10$ as the headline value and report $k = 1$ and $k = 50$ as a fixed robustness grid; because no timestamped preregistration is available, these choices are not described as preregistered.

All three scorers share one pre-specified convention (higher score, more out-of-distribution) and are evaluated by AUROC and FPR-at-95%-TPR on the same ISIC-test-vs-PAD-UFES split. To separate operational score direction from the presence of any scalar separation signal, we report two summaries: the primary *directed OOD AUROC*,

$$\text{AUROC}_{\text{dir}} = \Pr[s(X_{\text{OOD}}) > s(X_{\text{ID}})], \quad (1)$$

computed under that convention, and the descriptive *orientation-free separability*,

$$\text{AUROC}_{\text{sep}} = \max\{\text{AUROC}_{\text{dir}}, 1 - \text{AUROC}_{\text{dir}}\}. \quad (2)$$

The first tests whether the scorer is operationally valid in its intended direction; the second tests whether the scalar score retains any rank-based separation once direction is ignored. $\text{AUROC}_{\text{sep}}$ is a diagnostic, not the performance of a corrected detector: reversing polarity after observing evaluation labels is post-hoc and cannot establish deployable calibration. Because every directed AUROC observed here is below 0.5, $\text{AUROC}_{\text{sep}} = 1 - \text{AUROC}_{\text{dir}}$ throughout Section 4.4.

3.4 Domain-information probe

To test whether domain-discriminative information is present in an embedding independent of whether any distance-based scorer can access it, we train three supervised classifiers — logistic regression, a linear support vector machine, and a random forest, each at library-default hyperparameters with no tuning — to predict domain membership (ISIC-test vs. PAD-UFES) directly from the same z_{lesion} embeddings scored in Section 3.3. Each is a fresh classifier fit only for this analysis, with no access to the original training objective. We report 5-fold stratified cross-validated AUROC on out-of-fold predictions, avoiding the fit-and-evaluate-on-the-same-data leakage that would otherwise inflate apparent decodability at this sample size and dimensionality; AUROC is the primary probe metric because the split is class-imbalanced (approximately 2.2:1) and AUROC is threshold-independent and far less sensitive to that imbalance than accuracy, though we did not supplement it with balanced accuracy or precision-recall AUC. Probe AUROC is evidence of what is linearly and nonlinearly recoverable from the representation, not a claim about the training objective’s internal computation (Hewitt & Liang, 2019).

3.5 Statistical testing

The ladder has only three ordinal levels with unequal per-level seed counts (5, 5, 3). We test association using Kendall’s τ and, for ordered-group trend testing, the Jonckheere–Terpstra statistic (Terpstra, 1952; Jonckheere, 1954). For the ordered-ladder design, exact probabilities are computed over all distinct label arrangements consistent with the true per-rung seed counts; for the continuous–continuous design, the exact null over all $13!$ rank orderings is evaluated by its combinatorial distribution rather than by materializing 6.2 billion permutations. Both follow standard permutation-test principles (Good, 2005) and avoid an unverified asymptotic approximation at this sample size.

Given the limited number of independent checkpoints ($n = 13$), we emphasize consistency across analyses rather than significance from any single test; unadjusted exact p -values are reported without family-wise correction. The analysis used a criterion of $|\tau| \geq 0.3$ with significance at $\alpha = 0.05$ and a sign stable between the full ladder and the common-seed subset. Because no timestamped preregistration is available for this criterion, we do not characterize it as preregistered; Section 3.6 treats it as an analysis criterion and states what it can and cannot establish at these sample sizes.

Kendall’s τ p -values are two-sided throughout. Jonckheere–Terpstra p -values are one-sided upper-tail, the conventional reading of an ordered-alternative trend test: the statistic is centered at a positive null mean and is never negative, so a two-sided rule applied to it is not the same object as the two-sided Kendall test. The two are also not independent evidence at these designs. With a tie-free response they satisfy the exact identity $\tau_b = (2J - M)/\sqrt{Mn_0}$, where M is the number of cross-rung pairs and n_0 the total number of pairs, so the two-sided Jonckheere–Terpstra p -value is identical to the Kendall p -value at every design used here. Where both are reported below, they are one result expressed twice rather than mutual corroboration.

Table 2: Representation geometry across the orthogonality-strength ladder within the domain-adversarial regime (mean $\pm$ SD).

Metric	runA_gr1 ($\lambda_{\text{orth}} = 0$)	runB_orth1 ($\lambda_{\text{orth}} = 1$)	runB ($\lambda_{\text{orth}} = 5$)
Condition number	75.5 $\pm$ 7.8	559.8 $\pm$ 125.9	5329.9 $\pm$ 2672.2
Fisher ratio (Hotelling–Lawley, $\times 10^5$)	5.41 $\pm$ 0.81	5.71 $\pm$ 0.43	5.63 $\pm$ 0.50
Fisher ratio (decoupled scalar)	4.69 $\pm$ 0.65	4.94 $\pm$ 0.37	4.95 $\pm$ 0.38
Mardia’s kurtosis (b)	739.9 $\pm$ 86.5	788.0 $\pm$ 240.9	679.9 $\pm$ 63.7
Mardia’s kurtosis (z)	341.1 $\pm$ 99.3	397.3 $\pm$ 257.6	270.7 $\pm$ 53.4
n seeds	5	5	3

3.6 Detectability, power, and interval estimates

A p -value above α states only that the observed statistic was not extreme under the null; it does not say which effect sizes the design could have detected, nor which the data exclude. We therefore report three further quantities for every association tested, all computed from the per-checkpoint results already collected, with nothing retrained. First, τ_{crit} : the smallest $|\tau|$ attainable at a design whose two-sided exact permutation p -value reaches 0.05, a property of n , the group sizes, and the tie structure alone. Second, simulation power (2×10^4 datasets per grid point, common random numbers across the grid) under alternatives parameterized by the expected reported τ . Third, 95% bias-corrected and accelerated (BCa) bootstrap intervals (Efron, 1987) over 2×10^4 within-rung resamples, so that the fixed 5/5/3 design is preserved. For an estimate at the design ceiling a conventional two-sided BCa interval is not reported, because the bootstrap distribution is necessarily one-sided and boundary-constrained; the exact permutation result remains the primary inference there. Appendix C gives full definitions, seeds, and results.

3.7 Reproducibility and availability

Every checkpoint, seed, and preprocessing parameter is recorded by explicit value or file path; no result depends on a directory-resolved or otherwise ambiguously selected checkpoint. Both datasets are public: ISIC 2019 through the ISIC Archive and PAD-UFES-20 through its Mendeley Data record. The anonymized supplement accompanying this submission contains the full experiment pipeline — embedding extraction from checkpoints, geometry diagnostics, the three distance scorers, the domain probes, and every statistical analysis in this paper — together with the random-seed manifest, the 13-row checkpoint-level results matrix from which the interval table and Figure 7 are recomputed, and one-command reproduction instructions. Trained checkpoints and the intermediate embeddings are too large for the supplement and will be released with the de-anonymized archival version.

4 Results

4.1 Covariance conditioning varies across the orthogonality-strength ladder

Condition number increased monotonically with λ_{orth} : 75.5 $\pm$ 7.8 (runA_gr1), 559.8 $\pm$ 125.9 (runB_orth1), 5329.9 $\pm$ 2672.2 (runB) — a two-order-of-magnitude change across the ladder (Table 2, Figure 1). The trend was significant by exact-permutation Kendall’s τ ($\tau = 0.84$, $p = 2.8 \times 10^{-5}$, $n = 13$), sign-stable on the common-seed subset ($\tau = 0.87$, $p = 0.0012$, $n = 9$), and confirmed by Jonckheere–Terpstra trend testing ($J = 55.0$ against a null mean of 27.5, exact one-sided $p = 1.4 \times 10^{-5}$; by the identity in Section 3.5 its two-sided counterpart is the Kendall p above, not independent corroboration). The observed $\tau = 0.84$ is the largest value the 5/5/3 design admits — a perfectly monotone ladder — and is the only association in this study that clears the design’s critical value of 0.473 (Section 3.6).

The remaining four geometry metrics showed no significant trend: Fisher ratio (Hotelling–Lawley), $\tau = 0.17$, $p = 0.52$; Fisher ratio (decoupled scalar), $\tau = 0.08$, $p = 0.80$; Mardia’s kurtosis (b), $\tau = -0.14$, $p = 0.61$; Mardia’s kurtosis (z), $\tau = -0.17$, $p = 0.52$ (Table 2).

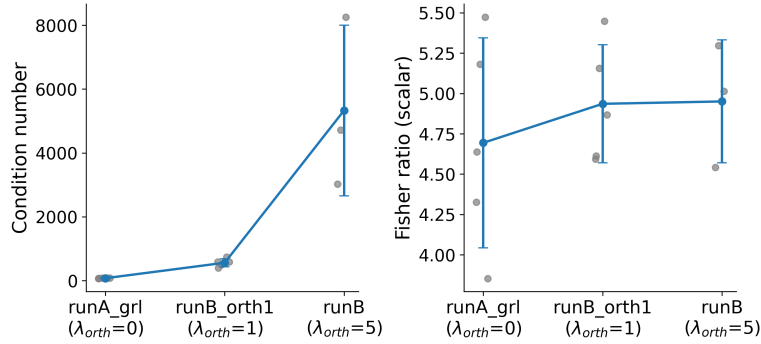


Figure 1: Representation geometry across the orthogonality-strength ladder. Condition number (left) and the decoupled Fisher-ratio scalar $\text{tr}(S_B)/\text{tr}(S_W)$ (right) for each of 13 checkpoints (gray points; mean $\pm$ SD in blue), across three levels of orthogonality strength. Condition number increases by approximately two orders of magnitude with λ_{orth} (Kendall’s $\tau = 0.84$, exact $p = 2.8 \times 10^{-5}$; Table 2); the Fisher-ratio scalar shows no significant trend.

4.2 Representation geometry shows no measurable association with Mahalanobis AUROC

Mahalanobis AUROC (ISIC-test vs. PAD-UFES) was 0.401 ± 0.028 (runA_gr1), 0.408 ± 0.022 (runB_orth1), and 0.393 ± 0.025 (runB) — below the chance value of 0.5 at every rung (Table 5 in Appendix B).

None of the five geometry metrics, including condition number, showed a significant association with Mahalanobis AUROC: condition number, $\tau = -0.13$, $p = 0.59$; Fisher ratio (HL), $\tau = -0.28$, $p = 0.20$; Fisher ratio (scalar), $\tau = -0.15$, $p = 0.51$; Mardia’s kurtosis (b), $\tau = -0.13$, $p = 0.59$; Mardia’s kurtosis (z), $\tau = -0.13$, $p = 0.59$ (Figure 4). AUROC itself showed no significant trend across the ladder (Jonckheere–Terpstra, $J = 25.0$, null mean 27.5, exact one-sided $p = 0.65$). Given only $n = 13$ checkpoints these results are inconclusive with respect to small-to-moderate monotonic associations rather than evidence of their absence: at this design no $|\tau|$ below 0.436 can reach $p \leq 0.05$ under the exact permutation null, and all five bootstrap intervals remain wide enough to contain $|\tau| = 0.3$, the widest admitting $|\tau| = 0.75$ (Section 3.6; Appendix C, Table 7).

An approximately two-orders-of-magnitude change in condition number (Section 4.1) is not accompanied by a significant change in Mahalanobis AUROC; no geometry metric tested shows a significant association with AUROC.

4.3 Distance-based scores are systematically lower for the target domain

Median Mahalanobis squared distance was lower for PAD-UFES than for ISIC-test at every one of the 13 checkpoints. Pooled across seeds within each rung: runA_gr1, ISIC-test median 13.50 vs. PAD-UFES median 8.57 ($n_{id} = 25335$, $n_{ood} = 11490$); runB_orth1, 14.10 vs. 8.99 (same n); runB, 13.20 vs. 8.08 ($n_{id} = 15201$, $n_{ood} = 6894$). Two candidate explanations for that gap — a feature-norm difference between domains, and excess attraction of PAD-UFES samples to the majority ISIC class — were examined and neither accounts for it on its own; Appendix B reports both analyses and Figure 5.

4.4 Directional misalignment is consistent across three distance-based scorer families

Pooled across all 13 checkpoints, AUROC_{dir} was 0.402 ± 0.024 (Mahalanobis), 0.418 ± 0.022 (cosine-to-centroid), 0.424 ± 0.023 (k -NN, $k = 1$), 0.414 ± 0.024 (k -NN, $k = 10$), and 0.411 ± 0.023 (k -NN, $k = 50$) (Table 3; Figure 6 in Appendix B). All five directed values fall within a 0.40–0.42 range, below the chance value of 0.5; the corresponding AUROC_{sep} values are 0.576–0.598. The scalar distances therefore retain weak domain-separation signal but consistently order PAD-UFES samples opposite to the scorers’ pre-specified OOD semantics. Despite substantially different scoring formulations — parametric with covariance,

Table 3: Directed OOD AUROC and orientation-free separability (ISIC-test vs. PAD-UFES) for five distance-based scorers (mean $\pm$ SD across seeds). Panel A preserves the pre-specified higher-distance-is-more-OOD direction by rung. Panel B separates this operational result from descriptive, orientation-free scalar separability.

Panel A: AUROC_{dir} by ladder rung			
Scorer	runA_gr1	runB_orth1	runB
Mahalanobis	0.401 $\pm$ 0.028	0.408 $\pm$ 0.022	0.393 $\pm$ 0.025
Cosine-to-centroid	0.411 $\pm$ 0.020	0.413 $\pm$ 0.015	0.439 $\pm$ 0.027
k -NN ($k = 1$)	0.424 $\pm$ 0.028	0.428 $\pm$ 0.020	0.416 $\pm$ 0.028
k -NN ($k = 10$)	0.413 $\pm$ 0.029	0.417 $\pm$ 0.020	0.408 $\pm$ 0.029
k -NN ($k = 50$)	0.411 $\pm$ 0.027	0.414 $\pm$ 0.020	0.408 $\pm$ 0.030
Panel B: pooled directional validity vs. orientation-free separability ($n = 13$)			
Scorer	AUROC _{dir}	AUROC _{sep}	Interpretation
Mahalanobis	0.402 $\pm$ 0.024	0.598 $\pm$ 0.024	reversed, weak
Cosine-to-centroid	0.418 $\pm$ 0.022	0.582 $\pm$ 0.022	reversed, weak
k -NN ($k = 1$)	0.424 $\pm$ 0.023	0.576 $\pm$ 0.023	reversed, weak
k -NN ($k = 10$)	0.414 $\pm$ 0.024	0.586 $\pm$ 0.024	reversed, weak
k -NN ($k = 50$)	0.411 $\pm$ 0.023	0.589 $\pm$ 0.023	reversed, weak

parametric without covariance, and non-parametric — all three families produced highly similar directional misalignment.

None of the five variants showed a significant association with λ_{orth} : Mahalanobis, $\tau = -0.08$, $p = 0.80$; cosine, $\tau = 0.32$, $p = 0.19$; k -NN ($k = 1$), $\tau = -0.05$, $p = 0.90$; k -NN ($k = 10$), $\tau = 0.02$, $p = 1.00$; k -NN ($k = 50$), $\tau = 0.08$, $p = 0.80$. Jonckheere–Terpstra gave the same result for all five (exact one-sided p between 0.10 and 0.65; not independent of the Kendall tests, by the identity in Section 3.5). As in Section 4.2, the small sample leaves these estimates inconclusive with respect to small-to-moderate effects: the critical value on the 5/5/3 ladder is $|\tau| = 0.473$ and all five intervals admit $|\tau| = 0.3$ (Appendix C, Table 7). Cosine-to-centroid has the largest point estimate ($\tau = 0.32$, 95% CI $[-0.21, 0.69]$); it exceeds 0.3 in magnitude but falls well short of significance at this design, and we do not read it as a trend.

These results do not show that the distance scores are devoid of domain-separation information. All five variants retain weak orientation-free separation and impose it in the direction opposite to their intended operational semantics. Because the reverse orientation is identified using evaluation labels, it is interpreted diagnostically and is not presented as a post-hoc corrected OOD detector.

Three structurally distinct distance-based scorer families report directed AUROC in the same 0.40–0.42 range and orientation-free separability of only 0.58–0.60. The scores are weakly separating but consistently misaligned with the pre-specified higher-distance-is-more-OOD direction; none shows a significant association with λ_{orth} .

4.5 Domain information remains decodable despite directionally misaligned distance scoring

On the identical embeddings scored in Section 3.3, three supervised probes recovered domain membership (ISIC-test vs. PAD-UFES) at: runA_gr1, 0.719 (logistic regression), 0.717 (linear SVM), 0.807 (random forest); runB_orth1, 0.741, 0.740, 0.815; runB, 0.750, 0.751, 0.800 (Table 4, Figure 2). AUROC exceeded 0.70 for all three probe families at every rung. None showed a significant association with λ_{orth} (logistic regression, $\tau = 0.26$, $p = 0.30$; linear SVM, $\tau = 0.26$, $p = 0.30$; random forest, $\tau = -0.11$, $p = 0.70$); these estimates are subject to the same $n = 13$ limitation, and each of the three intervals admits $|\tau| = 0.3$ (Appendix C, Table 7).

Table 4: Mean directed distance-scorer AUROC vs. domain-probe AUROC, by rung. Directed AUROC is the unweighted mean across the five scorer variants in Table 3. **baseline_soft** is a descriptive reference, not a fourth ladder point (Section 3.1).

Rung	Role	Directed AUROC	Probe (LR)	Probe (SVM)	Probe (RF)
runA_grl	primary ladder	0.412	0.719	0.717	0.807
runB_orth1	primary ladder	0.414	0.741	0.740	0.815
runB	primary ladder	0.413	0.750	0.751	0.800
baseline_soft	categorical reference ($n = 1$)	0.828	0.998	0.997	0.977

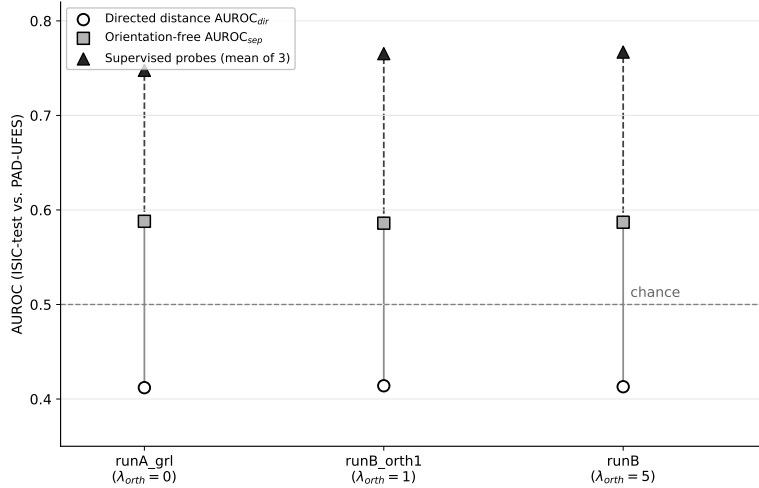


Figure 2: Domain decodability exceeds both directed distance ranking and orientation-free scalar separability. For each rung, the mean directed AUROC across five distance scorers (open circles) is linked to its orientation-free counterpart, $1 - \text{AUROC}_{\text{dir}}$ in this below-chance regime (gray squares), and to mean AUROC across three supervised domain probes (black triangles). Directed distance ranking remains near 0.41, orientation-free separability near 0.59, and supervised probe AUROC near 0.75–0.77. Solid segments mark the score-orientation gap; dashed segments mark the remaining gap between scalar distance separability and supervised decodability from the full embedding. Orientation-free values are diagnostic and do not represent a detector whose polarity was calibrated on independent data.

Three independent classifiers, applied to the same embeddings scored by the distance-based scorers in Section 3.3, recover domain membership at 0.72–0.81 AUROC at every λ_{orth} level tested; probe AUROC shows no significant association with λ_{orth} .

4.6 Descriptive reference: a conventional non-adversarial baseline

For a single ResNet-50 checkpoint (**baseline_soft**; one seed, architecture and representation dimensionality differing from the ladder, Section 3.1), pooled distance-scorer AUROC was 0.828 and probe AUROC was 0.998 (logistic regression), 0.997 (linear SVM), and 0.977 (random forest) (Table 4).

5 Discussion

5.1 Interpretation

The results support a three-level interpretation that separates operational validity, scalar-score information, and information recoverable from the full embedding.

Level 1: the pre-specified operational direction fails. All five distance-score variants produced $\text{AUROC}_{\text{dir}} = 0.402\text{--}0.424$ under the higher-distance-is-more-OOD convention (Section 4.4), ranking PAD-UFES samples as more source-typical than ISIC-test samples. This is an operational polarity failure: a reliability gate using the direction specified before evaluation would rank the audited target domain incorrectly. Agreement across Mahalanobis, cosine-to-centroid, and pooled k -NN shows the pattern is not confined to covariance inversion or to one value of k .

Level 2: the scalar scores retain weak separation signal. A directed AUROC below 0.5 does not imply absence of information. Removing orientation yields $\text{AUROC}_{\text{sep}} = 0.576\text{--}0.598$, so the distances weakly separate the two domains while ordering them opposite to the intended semantics — neither chance-level non-separation nor a valid detector obtained by negation. The reverse orientation was identified from evaluation labels; without polarity selection on independent calibration data, $\text{AUROC}_{\text{sep}}$ remains diagnostic rather than deployable performance.

Level 3: the full embeddings support stronger supervised decodability. Probes recovered domain membership at 0.72–0.81 AUROC from the same embeddings (Section 4.5). Figure 2 therefore exposes two distinct gaps: between the intended and observed direction of the scalar scores, and between their weak orientation-free separability and the stronger decodability available from the full representation. Probe success is an existence test only; it does not imply that the original classifier or any unsupervised reliability mechanism uses the decoded information.

This also constrains the geometry result. Covariance conditioning moved by approximately two orders of magnitude (Section 4.1), but neither directed Mahalanobis AUROC nor the other scorer families tracked that change. The conclusion is not that geometry is irrelevant, but that condition number alone is insufficient to predict either the direction or the strength of the rank ordering these source-fitted scores induce. Section 4.3 confirms a systematic distance-ordering difference without fully explaining its origin; establishing why the targeted domain is ranked as source-typical remains future work. The supported claim is therefore scoped: within this domain-adversarial regime and for its adversarial target domain, the tested source-fitted distance rules are directionally misaligned, their scalar outputs retain weak separation, and the full embeddings retain stronger supervised domain decodability. This is not a claim that scalar distances contain no information, that supervised probes constitute an OOD detector, or that domain-adversarial training or orthogonality regularization causes the observed ordering.

5.2 Scorer-relevant geometry

Part of why a large conditioning change need not move AUROC is structural rather than empirical: for a full-rank affine change of coordinates the unregularized squared Mahalanobis distance is invariant while the condition number $\kappa(\Sigma)$ can change by orders of magnitude, and cosine-to-centroid, k -NN, and AUROC itself each preserve a different set of transformations. The chain from raw geometric change to score change to ranking change to AUROC change is therefore not automatic, and “geometry” is not one object for this purpose. Appendix D develops this argument, states each scorer’s invariances, and draws out the audit principle that follows: evaluation should separately test score polarity, orientation-free separation, and information recoverable from the representation, while identifying the invariances of the downstream scorer.

5.3 Relation to literature

Mahalanobis-distance OOD detection (Lee et al., 2018) and its non-parametric successors, including pooled k -nearest-neighbor scoring (Sun et al., 2022), are typically evaluated by benchmarking AUROC across datasets and architectures, not by separately auditing score polarity, orientation-free separability, and full-embedding decodability after a representation-changing intervention. This study proposes no competitor to either; both are used as audit instruments, alongside cosine-to-centroid scoring. Their common result here is a shared directional misalignment with weak residual separation — not a comparative ranking and not evidence that their scalar outputs contain no information.

Domain-adversarial representation learning (Ganin & Lempitsky, 2015; Ganin et al., 2016) and orthogonality-based shared/private separation (Bousmalis et al., 2016) are among the strategies proposed to reduce reliance on nuisance or shortcut features (Geirhos et al., 2020). This study does not identify that training family as the cause of the observed ordering, because every ladder checkpoint already uses the domain-adversarial regime and the conventional baseline is not matched. It asks instead whether source-fitted distance scores retain their intended polarity and how much separation survives when orientation is ignored; for the audited target domain the intended polarity fails and residual separability is weak, while causal attribution to domain-adversarial training, orthogonality regularization, architecture, or dimensionality remains outside the evidence provided here. Appendix E.7 extends this to the probing and geometry–performance literatures and to the practitioner implications.

5.4 Limitations

PAD-UFES is not an arbitrary target domain: it is the domain the architecture’s adversarial branch is trained against. Our conclusions therefore characterize representations learned under this training regime, not all domain shifts.

AUROC_{sep} is computed from the observed evaluation ordering and is reported only as an orientation-free diagnostic. This study includes no independent calibration split on which score polarity could be selected and then transferred unchanged to a held-out target. It therefore does not establish that negating a distance score yields an operational detector, that the reverse polarity transfers to another shift, or that a deployed system could identify the required polarity without labeled shifted data.

The conventional non-adversarial baseline (Section 4.6) differs from the ladder in architecture, representation dimensionality, and training recipe simultaneously, and is a single checkpoint. It is descriptive and supports no causal attribution of the ladder findings to domain-adversarial training or orthogonality regularization as opposed to architecture or dimensionality; a matched non-adversarial checkpoint at the same architecture and 16-dimensional representation would be needed, and none exists in the available inventory.

The ladder spans three ordinal levels with unequal seed counts ($n = 13$; $n = 9$ on the common-seed subset). The exact critical values are 0.436, 0.473, and 0.609, while power at an expected $\tau = 0.3$ is only 0.27, 0.23, and 0.14 (Appendix C). The combined analysis criterion — $|\tau| \geq 0.3$ and $p \leq 0.05$ — could therefore not be satisfied by an association of exactly that magnitude at any of the three designs, and none of the seventeen non-significant intervals excludes a moderate association of magnitude 0.3 in at least one direction. The consistent point-estimate pattern bounds but does not eliminate uncertainty about whether λ_{orth} modulates the observed misalignment; the tests are also correlated, reusing metrics, scorers, and probes derived from the same embeddings. Exact-permutation testing was used throughout in place of unverified asymptotic approximations, and p -values carry no family-wise correction; a reader applying a stricter correction should note that the one large, load-bearing effect — condition number against λ_{orth} , $\tau = 0.840$, the design maximum — survives any standard correction, while every non-significant result was reported as such before any correction was considered.

This study includes no replication on a domain independent of the ISIC/PAD-UFES pair and no check of generalization beyond the single audited architecture. Both were considered and omitted: a third-domain replication needs a dataset without documented image overlap with the training data, which was unavailable within scope, and the cross-architecture check was not pursued for the same reason. Neither gap is filled by the baseline reference, which addresses architecture and dimensionality but not domain independence. The bootstrap check that would quantify shared estimation noise between condition number and Fisher ratio, both derived from the same fitted precision matrix, was not performed; condition number, not Fisher ratio, drives the geometric finding in Section 4.1, which bounds but does not remove this uncertainty. Mardia’s kurtosis null was calibrated with 200 bootstrap resamples per checkpoint; the regularization constant ($\varepsilon = 10^{-5}$) and the 16-dimensional representation were inherited from the audited classifier rather than varied here. “Reliability estimation” throughout means out-of-distribution detection AUROC specifically; calibration (Guo et al., 2017) and selective prediction, including distribution-free variants (El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017; Angelopoulos & Bates, 2023), are not addressed.

These limitations reinforce the three-level interpretation: the pre-specified operational direction is invalid for the audited target domain, the scalar scores retain weak orientation-free separation, and the full embeddings support stronger supervised decodability. None of these observations alone establishes transfer to an unseen domain or the performance of a calibrated deployment system.

6 Conclusion

This study audited the relationship among measured representation structure, distance-based OOD ordering, scalar separability, and supervised domain decodability within a fixed domain-adversarial training regime. Across an orthogonality-strength ladder, covariance conditioning changed by approximately two orders of magnitude while the other four geometry summaries showed no significant ladder trend, and that geometric change was not accompanied by a corresponding change in Mahalanobis OOD ranking or in any other tested distance-score family.

Under the pre-specified higher-distance-is-more-OOD convention, five variants from three scorer families produced $\text{AUROC}_{\text{dir}} = 0.402\text{--}0.424$. These scores were not devoid of domain-separation information: ignoring orientation yields $\text{AUROC}_{\text{sep}} = 0.576\text{--}0.598$. The information encoded in the scalar distances was therefore weak and systematically ordered opposite to the intended OOD semantics; because that reverse orientation was identified from evaluation labels, it is a diagnostic finding rather than an operationally corrected detector. Supervised probes on the full embeddings recovered substantially stronger domain membership information at $0.72\text{--}0.81$ AUROC.

The supported conclusion is narrower and more precise than either complete distance inaccessibility or a causal claim about disentanglement: within the audited representations, domain membership remains decodable, scalar distance scores retain weak separation, and the tested source-fitted distance rules impose a directionally misaligned OOD ordering. Raw geometric summaries, scorer-relevant geometry, score orientation, orientation-free separability, and supervised decodability should be treated as distinct objects of evaluation. Reliability mechanisms layered on learned representations should be re-validated after representation-changing interventions — including explicit validation of score polarity on data independent of the final evaluation set — rather than assumed to inherit validity from classification performance or generic geometric diagnostics.

Broader Impact Statement

This study is a retrospective, methodological audit conducted entirely on two pre-existing, publicly available, de-identified skin-lesion image datasets. It trains no new diagnostic classifier, deploys no system, and releases no model intended for clinical use; the audited checkpoints form an orthogonality-strength ladder within a domain-adversarial regime for the purpose of this analysis. No patients were recruited and no new clinical images were acquired.

The central risk this work is positioned to mitigate, rather than create, is the silent failure of a distance-based reliability safeguard in a deployed pipeline: if such a safeguard is assumed to function on representations produced by a domain-adversarial training regime, without being re-validated against the domain targeted during training, it may fail to flag inputs from precisely that domain. Making this failure mode explicit and measurable is intended to reduce, not increase, the risk of over-trusting a distance-based safeguard in a safety-critical medical imaging pipeline. We are not aware of a plausible negative use of this study’s findings or artifacts: the audit protocol, the geometry metrics, and the statistical tests reported here characterize an existing training regime rather than introduce a new capability that could be misapplied. We therefore do not identify a significant risk of harm requiring further mitigation beyond the cautions already stated in Section 5.4 regarding the scope of the claims.

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Declarations

Ethical approval. This study used publicly available de-identified datasets; therefore, ethical approval was not required.

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CRedit authorship contribution statement. D.-V. Tran: conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, visualization, writing—original draft. Q.-T. Huynh: methodology, formal analysis, writing—review and editing.

Use of Generative AI

Generative AI tools were used to assist with language editing, structural reorganisation, and improvement of manuscript readability, including LaTeX formatting and reference formatting. All scientific content, study design, experiments, analyses, interpretations, and final approval of the manuscript were performed and verified by the authors, who have reviewed and edited the output and take full responsibility for the content of this publication.

Data Availability

Both datasets used in this study are publicly available. The ISIC 2018 dataset is accessible through the ISIC Archive (<https://challenge.isic-archive.com/data/>). The PAD-UFES-20 dataset is available at <https://data.mendeley.com/datasets/zr7vgbcyr2>. The submission supplement contains fixed random seeds, the power-analysis script, derived CSV inputs used for the bootstrap-interval table and Figure 7, and reproduction instructions. The original 13-row checkpoint-level output matrix required to recompute the BCa endpoints, the full set of derived result tables, the extracted representations, and the per-sample distances are archived at <https://doi.org/10.5281/zenodo.22130029>. The analysis code is additionally maintained at <https://github.com/TranDucVinh-20225538/paper-3>. Model checkpoints are available from the corresponding author on request.

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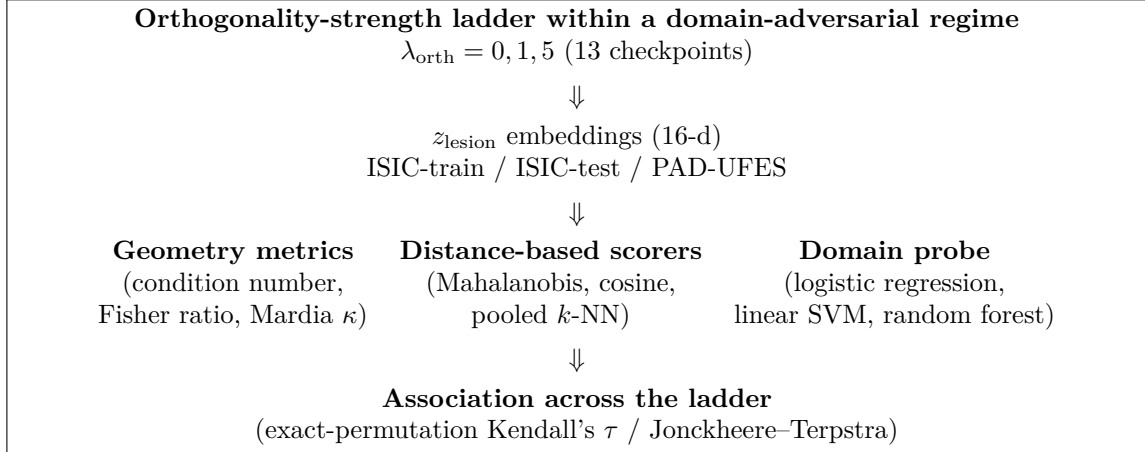


Figure 3: Study design. An orthogonality-strength ladder within a domain-adversarial regime comprises three checkpoint families ($\lambda_{\text{orth}} = 0, 1, 5$) sharing architecture and representation dimensionality. It is used to test whether an implicit assumption—that representation geometry reflects reliability-estimation quality—holds across changes in orthogonality strength within that regime. Geometry metrics, three distance-based scorers, and domain-information probes are computed on identical embeddings for each of 13 checkpoints (Section 3.1).

A Extended methods

This appendix holds the pipeline diagram and the full metric definitions that Section 3 states in summary form. Nothing here differs from what was computed; it is the same protocol at greater length.

A.1 Representation geometry metrics

The selected metrics were pre-specified to represent three complementary aspects of representation geometry – covariance conditioning, class separation, and multivariate normality – rather than to maximize empirical correlation with downstream performance.

- **Condition number:** the ratio of the largest to smallest eigenvalue of the regularized, shared within-class covariance matrix used to fit the Mahalanobis estimator (Section 3.3), measuring how numerically stable that covariance’s inversion is.
- **Fisher ratio:** the Hotelling–Lawley trace, $\text{tr}(\Sigma^{-1}S_B)$, computed from the same fitted precision matrix, measuring class separation relative to within-class spread. We additionally report a decoupled scalar companion, $\text{tr}(S_B)/\text{tr}(S_W)$, which involves no matrix inverse and is therefore not entangled with condition number’s own estimation noise – the two are reported together because they are not statistically independent measurements when both derive from the same Σ^{-1} .
- **Mardia’s multivariate kurtosis** (Mardia, 1970): quantifies departure from the class-conditional Gaussian shape the Mahalanobis estimator assumes. The classical closed-form asymptotic null does not apply once class means and covariance are estimated in-sample from the same residuals being tested; its null distribution is calibrated by parametric bootstrap (200 resamples per checkpoint) rather than the textbook formula.

All three metrics are computed on the identical fitted Mahalanobis parameters used for scoring (Section 3.3), not from an independently re-estimated covariance, so that geometry and reliability estimation describe the same fitted object.

Table 5: Mahalanobis-distance AUROC (ISIC-test vs. PAD-UFES), by rung (mean $\pm$ SD across seeds).

Rung	λ_{orth}	n seeds	AUROC	FPR@95%TPR
runA_grl	0	5	0.401 ± 0.028	0.985 ± 0.009
runB_orth1	1	5	0.408 ± 0.022	0.980 ± 0.012
runB	5	3	0.393 ± 0.025	0.985 ± 0.003

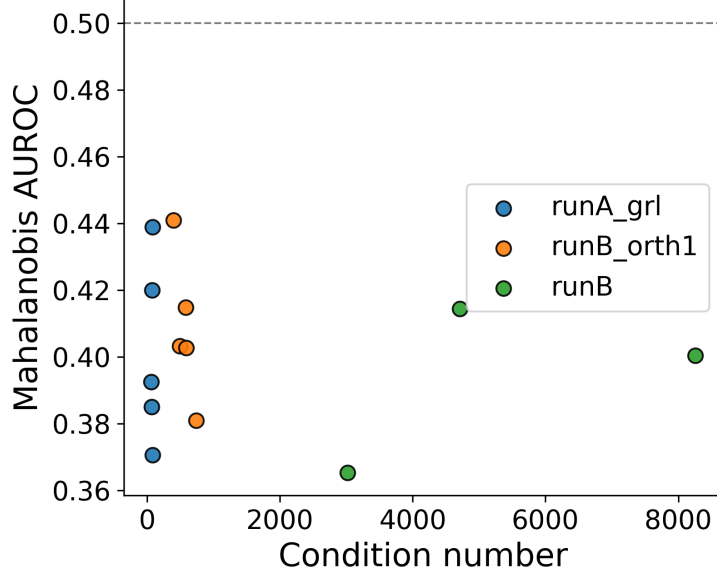


Figure 4: Representation geometry does not predict Mahalanobis AUROC. Condition number vs. Mahalanobis-distance AUROC (ISIC-test vs. PAD-UFES), one point per checkpoint, colored by rung. No association is apparent (Kendall’s $\tau = -0.13$, $p = 0.59$; Table 2); the remaining four geometry metrics show the same pattern.

B Supplementary results

Section 4.2 reports Mahalanobis AUROC by rung and its association with each geometry metric; Table 5 and Figure 4 give those results in full.

B.1 Two candidate explanations for the distance gap

Section 4.3 reports that median Mahalanobis distance is lower for PAD-UFES than for ISIC-test at every checkpoint, and notes that two obvious explanations do not account for it on their own. Those two analyses are reported here.

Feature-norm distributions differed modestly between domains. Pooled median $\|z\|$: **runA_grl**, ISIC-test 6.31 vs. PAD-UFES 6.21; **runB_orth1**, 6.38 vs. 6.28; **runB**, 6.28 vs. 5.97. At the level of pooled means, the direction was not consistent across rungs (**runA_grl** mean: 6.72 ISIC-test vs. 6.83 PAD-UFES). The direction of this difference was not consistent at the level of means.

Nearest-centroid classification assigned PAD-UFES samples to the majority ISIC class (Nevus) at 61.6% (**runA_grl**), 61.9% (**runB_orth1**), and 63.9% (**runB**), compared to 53.0%, 53.1%, and 53.4% respectively for ISIC-test samples classified against their own nearest centroid. This was 8.7–10.5 percentage points higher than the ISIC-test baseline rate (ratio 1.16–1.20).

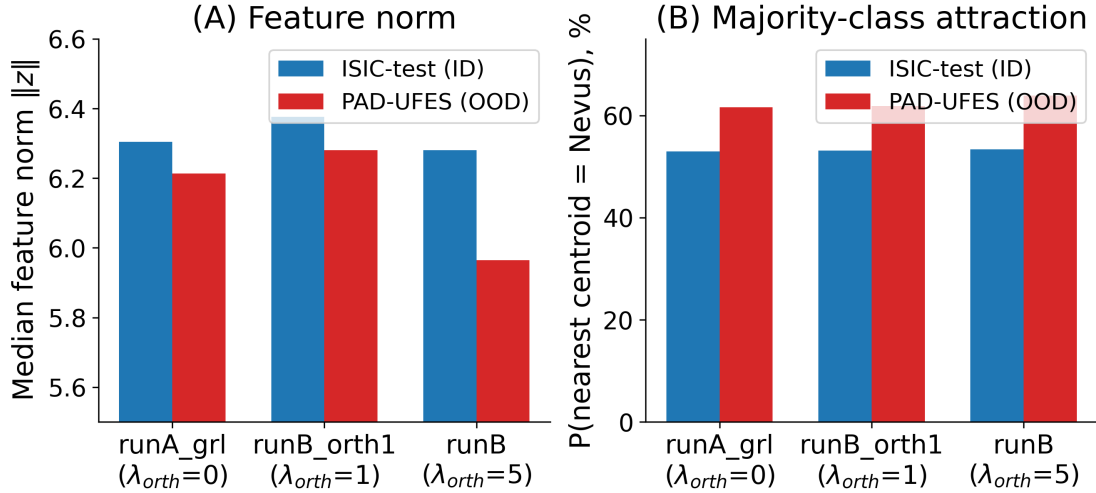


Figure 5: Neither feature-norm difference nor majority-class attraction fully accounts for the raw ID/OOD distance gap. (A) Pooled median feature norm, ISIC-test vs. PAD-UFES, per rung. (B) Nearest-centroid assignment of ISIC-test and PAD-UFES samples to the majority ISIC class (Nevus), per rung. The norm difference is modest and not consistent in direction at the level of means; PAD-UFES’s excess attraction to the majority class (8.7–10.5 percentage points above the ISIC-test baseline) is real but does not, on its own, account for the full distance separation in Figure 4.

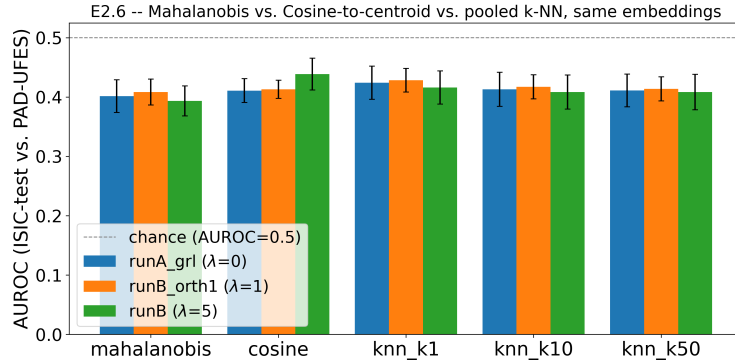


Figure 6: Three distance-based scorer families are directionally misaligned alike. Directed OOD AUROC (ISIC-test vs. PAD-UFES) for Mahalanobis distance, cosine-to-centroid similarity, and pooled k -nearest-neighbor distance ($k = 1, 10, 50$), computed on the identical embeddings and grouped by rung (mean $\pm$ SD across seeds). Dashed line: chance (AUROC = 0.5). All five variants fall within 0.40–0.42 under the pre-specified higher-distance-is-more-OOD convention; their pooled orientation-free separability is 0.576–0.598 (Table 3, Panel B).

B.2 Per-rung view of the five scorer variants

Figure 6 plots the values tabulated in Table 3, Panel A.

C Detectability, power, and interval estimates

This appendix reports the three quantities defined in Section 3.6 for every association tested in this paper. All of them are computed from the per-checkpoint results already reported in Section 4: no checkpoint is re-evaluated, no model is retrained, and no additional data are used.

Table 6: Detectability of the two permutation designs used in this study. Design A is the continuous–continuous test of a geometry metric against Mahalanobis AUROC (Section 4.2); Design B is the ordered ladder used for every test against λ_{orth} (Sections 4.1, 4.4 and 4.5). “Null space” is the number of distinct label arrangements enumerated for the exact test. τ_{crit} is the smallest $|\tau|$ attainable at that design whose two-sided exact permutation p -value reaches 0.05: no dataset, however clean, can produce a significant association below it. Power is computed by simulation (2×10^4 datasets per grid point, common random numbers) under an alternative whose expected value of the reported statistic equals the stated τ — bivariate normal for Design A, a linear dose shift $y = \theta d + \varepsilon$ for Design B. $\text{MDE}_{80\%}$ is the expected τ at which power first reaches 0.80.

Design	n	Null space	Max. $ \tau $	τ_{crit}	Power at		$\text{MDE}_{80\%}$
					$\tau = 0.3$	$\tau = 0.5$	
A: continuous–continuous	13	13!	1.000	0.436	0.27	0.71	0.54
B: 5/5/3 ordered ladder	13	72,072	0.840	0.473	0.23	0.63	0.57
B: 3/3/3 common-seed subset	9	1,680	0.866	0.609	0.14	0.39	0.69

C.1 What the designs could have detected

Two permutation designs are used in this paper and they are not interchangeable. Design A is continuous–continuous (a geometry metric against Mahalanobis AUROC, $n = 13$, tie-free), and its exact null distribution is defined over all 13! rank orderings and computed combinatorially; the implementation does not sequentially materialize 6.2 billion permutations. Design B tests an ordinal dose against a value (a geometry metric, scorer AUROC, or probe AUROC against λ_{orth}), where the dose is tied by construction and the exact null runs over the 72,072 distinct label arrangements consistent with the 5/5/3 rung counts, or the 1,680 arrangements of the 3/3/3 common-seed subset. Table 6 gives τ_{crit} , the attainable maximum $|\tau|$, and power for each; Figure 7(A) shows the full power curves.

Design B’s attainable maximum is 0.840 rather than 1.000 because a tied predictor caps τ_b below unity: with 5/5/3 rung sizes, a perfectly monotone ladder — every checkpoint at a lower λ_{orth} ranked below every checkpoint at a higher one — yields exactly $\tau_b = 0.840$. The condition-number result of Section 4.1 attains that ceiling.

C.2 What the data exclude

Table 7 gives a 95% bias-corrected and accelerated (BCa) bootstrap interval (Efron, 1987) for every τ reported in the paper, over 2×10^4 resamples drawn *within* rung so that the 5/5/3 design is held fixed; λ_{orth} is a fixed design factor, not a sampled one, and resampling across rungs would bootstrap a quantity this experiment never estimated. Figure 7(B) plots the same intervals against the two designs’ critical values.

Resampling with replacement necessarily creates ties in a response that was tie-free as observed, and the two standard tie treatments pull an interval in opposite directions: the τ_b denominator $\sqrt{(n_0 - t_x)(n_0 - t_y)}$ shrinks as ties accumulate and so inflates $|\tau|$ on tied resamples, whereas the design-normalized form $(2J - M)/\sqrt{Mn_0}$ pins the denominator to the original design and cannot exceed the design ceiling. The two are identical on the observed data. Both were computed; over the tests reported here their endpoints differ by a median of 0.027 and at most 0.091, and the two agree on every $|\tau| = 0.3$ exclusion verdict, so no statement in this paper depends on the choice. Table 7 reports the τ_b version.

One case does not receive a conventional two-sided BCa interval. Where an observed τ attains the maximum its design permits — condition number against λ_{orth} , $\tau = 0.840$ — every resample lies at or below the observed value, the resampling distribution is one-sided and boundary-constrained by construction. A conventional two-sided BCa interval would therefore not have its usual interior-estimate interpretation. That case is reported as a point without such an interval, and its exact permutation p -value is the primary inference.

Table 7: Every Kendall’s τ reported in this paper with a 95% bootstrap confidence interval (2×10^4 stratified resamples, bias-corrected and accelerated; resampling is within-rung, so the 5/5/3 design is held fixed). “Largest $|\tau|$ not excluded” is the interval endpoint furthest from zero: the effect size each non-significant result still permits.

Test	τ	95% CI	Largest $ \tau $ not excluded
<i>Geometry metric vs. λ_{orth} (Section 4.1)</i>			
Condition number	0.840	at design ceiling — see note	
Fisher ratio (Hotelling–Lawley)	0.168	[−0.461, 0.620]	0.62
Fisher ratio (decoupled scalar)	0.076	[−0.532, 0.532]	0.53
Mardia’s kurtosis (b)	−0.137	[−0.400, 0.206]	0.40
Mardia’s kurtosis (z)	−0.168	[−0.412, 0.174]	0.41
<i>Geometry metric vs. Mahalanobis AUROC (Section 4.2)</i>			
Condition number	−0.128	[−0.507, 0.200]	0.51
Fisher ratio (Hotelling–Lawley)	−0.282	[−0.750, 0.183]	0.75
Fisher ratio (decoupled scalar)	−0.154	[−0.611, 0.324]	0.61
Mardia’s kurtosis (b)	−0.128	[−0.500, 0.233]	0.50
Mardia’s kurtosis (z)	−0.128	[−0.534, 0.216]	0.53
<i>Directed scorer AUROC vs. λ_{orth} (Section 4.4)</i>			
Mahalanobis	−0.076	[−0.524, 0.432]	0.52
Cosine-to-centroid	0.321	[−0.210, 0.688]	0.69
k -NN ($k = 1$)	−0.046	[−0.500, 0.556]	0.56
k -NN ($k = 10$)	0.015	[−0.564, 0.489]	0.56
k -NN ($k = 50$)	0.076	[−0.524, 0.545]	0.55
<i>Domain-probe AUROC vs. λ_{orth} (Section 4.5)</i>			
Logistic regression	0.260	[−0.210, 0.656]	0.66
Linear SVM	0.260	[−0.210, 0.656]	0.66
Random forest	−0.107	[−0.560, 0.368]	0.56

Note. Condition number vs. λ_{orth} attains $\tau = 0.840$, the maximum value the 5/5/3 design permits (a perfectly monotone ladder). Every bootstrap resample therefore lies at or below the observed value, and the resampling distribution is one-sided and boundary-constrained by construction. A conventional two-sided BCa interval is therefore not reported; the exact permutation p -value (2.8×10^{-5}) is the primary inference for that effect. Of the seventeen non-significant associations, none has an interval that excludes $|\tau| = 0.3$.

D Scorer invariance, sensitivity, and what “geometry” can predict

This appendix develops the argument summarized in Section 5.2.

The results are easier to interpret if raw representation geometry is distinguished from *scorer-relevant geometry*. For a full-rank affine change of coordinates $z' = Az + b$, with class means and the shared covariance transformed consistently as $\mu'_c = A\mu_c + b$ and $\Sigma' = A\Sigma A^\top$, the unregularized squared Mahalanobis distance is invariant:

$$(z' - \mu'_c)^\top (\Sigma')^{-1} (z' - \mu'_c) = (z - \mu_c)^\top \Sigma^{-1} (z - \mu_c). \quad (3)$$

Yet the numerical condition number $\kappa(\Sigma)$ can change greatly under a non-orthogonal A . Thus covariance conditioning can move by orders of magnitude without forcing Mahalanobis rankings, and therefore AUROC, to move commensurately. The small ridge term used here ($\varepsilon = 10^{-5}$) weakens exact affine invariance in finite samples but does not change the conceptual point: condition number is a property of a fitted covariance representation, not a sufficient statistic for Mahalanobis OOD ranking.

The other scorers preserve different structure. Cosine-to-centroid distance is invariant to a common orthogonal rotation and positive uniform rescaling, but is sensitive to translation and anisotropic transformations. Euclidean k -NN distance is invariant to translation and orthogonal transformations; positive uniform rescaling multiplies all distances by the same constant and therefore preserves their ordering, whereas anisotropic

transformations can change neighborhoods. Finally, AUROC itself is invariant to every strictly increasing transformation of a score. These invariances imply that the chain

$$\text{raw geometric change} \Rightarrow \text{score change} \Rightarrow \text{ranking change} \Rightarrow \text{AUROC change}$$

is not automatic.

This distinction changes the interpretation of the present null associations. The finding is not that “geometry does not matter” for distance-based reliability estimation. Rather, the measured covariance-conditioning change is not sufficient to predict directed OOD ranking in this regime, and the other measured summaries do not show systematic ladder trends. The convergence of Mahalanobis, cosine-to-centroid, and pooled k -NN on the same polarity reversal and similarly weak orientation-free separability points to a mismatch between the intended OOD semantics and the particular geometric orderings these scorers impose. This suggests a general audit principle beyond medical imaging: when representation-learning interventions are combined with post-hoc reliability mechanisms, evaluation should separately test score polarity, orientation-free separation, and information recoverable from the underlying representation, while identifying the invariances and sensitivities of the downstream scorer.

E Extended related work

Section 2 summarizes six adjacent literatures in four paragraphs. This appendix gives the extended treatment of each.

E.1 Post-hoc OOD Detection and Training-Time Enhancements

Post-hoc out-of-distribution (OOD) detection methods are typically grouped by whether they read the classifier’s output layer or its internal representation. Maximum softmax probability (Hendrycks & Gimpel, 2017) and its temperature-scaled and input-perturbed successor ODIN (Liang et al., 2018) score an input from the output logits directly; energy-based scoring (Liu et al., 2020) aggregates the full logit vector rather than its maximum; ReAct (Sun et al., 2021) rectifies unusually high activations before scoring; ViM (Wang et al., 2022) combines a logit-space term with a residual computed in feature space; and activation-shaping (Djurisic et al., 2023) prunes a large fraction of late-layer activations at inference time without using any training-set statistics. Complementary training-time approaches can instead reshape representations or logits so that subsequent post-hoc OOD scores become more effective. Extended Logit Normalization (ELoNorm) (Ding et al., 2026), for example, introduces a feature-distance-aware training objective designed to improve a range of post-hoc OOD detectors while preserving in-distribution classification performance. Distance-based scorers treat the penultimate-layer representation as the object of interest: Mahalanobis distance (Lee et al., 2018) fits class-conditional Gaussians and scores an input by its distance to the nearest class mean; pooled k -nearest-neighbor distance (Sun et al., 2022) replaces this parametric fit with a non-parametric density estimate; nearest-neighbor guidance (Park et al., 2023) instead uses a k -NN-derived correction term to reduce classifier overconfidence rather than to replace the classifier score outright; and Gram-matrix scoring (Sastry & Oore, 2020) extracts still other structural summaries of the same representation. Benchmarking across this family, surveyed for the convolutional setting by Yang et al. (2024) and, as the field has moved toward large pretrained encoders, for the vision-language and foundation-model setting by Miyai et al. (2025), typically compares detectors across architectures and datasets under a fixed, unperturbed training regime. This study instead holds architecture and dataset fixed and asks whether the same detector’s behavior on the same representation changes when the representation itself is deliberately reshaped by training – a question orthogonal to, and not addressed by, cross-detector benchmarking.

E.2 Domain-Adversarial Training and Representation Disentanglement

Domain-adversarial training (Ganin & Lempitsky, 2015; Ganin et al., 2016) learns a representation that a discriminator cannot use to identify the input’s domain, by reversing the discriminator’s gradient during backpropagation. Related approaches pursue a similar goal through different mechanisms: domain separation networks (Bousmalis et al., 2016) factor a representation into shared and domain-private subspaces

and use a difference loss that imposes a soft subspace-orthogonality constraint between them; correlation-alignment methods (Sun & Saenko, 2016) instead match second-order feature statistics across domains without an adversarial objective; and adversarial discriminative adaptation (Tzeng et al., 2017) applies the same discriminator-based principle asymmetrically across a source and target encoder. A related but distinct family, disentangled representation learning (Higgins et al., 2017), seeks to factor a representation into semantically separable generative factors rather than to suppress one nuisance factor specifically; Locatello et al. (2019) shows that such disentanglement is not identifiable from unlabeled data without further assumptions, a caution that does not directly bear on the supervised, single-nuisance-variable setting audited here but that motivates treating any claimed disentanglement as an empirical question rather than an assumption. All of these methods are evaluated in their originating literature primarily by downstream task accuracy or by a discriminator’s inability to recover the suppressed variable; none of the above is evaluated, to our knowledge, by whether a distance-based reliability estimator built on the resulting representation continues to function as it did before the intervention, which is the question this study takes up.

E.3 Shortcut Learning and Spurious Correlations

The failure mode motivating disentanglement – a classifier exploiting a nuisance correlate of the label rather than task-relevant structure – has been documented repeatedly in clinical imaging. Chest radiograph classifiers have been shown to encode hospital- and scanner-specific artefacts (Zech et al., 2018); COVID-19 detectors trained on heterogeneous radiograph sources were found to rely on scanner-specific cues detectable by saliency attribution rather than on pathology (DeGrave et al., 2021); and in dermatology specifically, surgical skin markings have been shown to inflate a model’s predicted probability of malignancy (Winkler et al., 2019), while Bissoto et al. (2020) report that ISIC-trained classifiers retain above-chance accuracy even when the lesion region itself is masked, implicating acquisition-context cues directly. Geirhos et al. (2020) synthesizes this literature under the term shortcut learning and argues that standard accuracy metrics provide no warning of it. This body of work establishes that shortcut reliance is a real and recurring failure mode and motivates disentanglement as a countermeasure; it does not, however, examine what such a countermeasure does to a reliability estimator layered on top of the resulting representation, which is the gap this study addresses.

E.4 Probing Classifiers and Representation Analysis

Linear probing – training a classifier on a frozen representation to measure what that representation linearly encodes – was introduced as a diagnostic for what intermediate layers of a network represent (Alain & Bengio, 2016) and has since been used across modalities to characterize learned representations independently of the objective that produced them (Hewitt & Liang, 2019). Belinkov (2022) surveys this literature and cautions specifically against over-interpreting a successful probe as evidence that the original model computes or uses the probed information in the way the probe accesses it. This study’s use of probing follows that caution directly: probe AUROC here is offered only as a measurement of what is linearly and non-linearly recoverable from an embedding, independent of any distance-based scorer, and is not offered as a claim about the disentanglement objective’s internal computation.

E.5 Representation Geometry and Estimator Performance

A separate line of work characterizes the geometry of learned representations directly, rather than what they linearly encode. Intrinsic dimensionality, estimated by nearest-neighbor methods such as the maximum-likelihood estimator of Levina & Bickel (2004), the two-nearest-neighbor estimator of Facco et al. (2017), or applied to deep network representations by Ansuini et al. (2019), is one such summary; the conditioning of the class-covariance matrix used to fit a Mahalanobis estimator is another. Across a range of frozen vision foundation-model backbones, Janiak et al. (2026) report that intrinsic dimensionality and within-class spectral decay track how well Mahalanobis-style detectors perform, characterizing this relationship as an association observed across backbones rather than as a causal claim. That association is measured between representations that differ simultaneously in pretraining corpus, objective, architecture, and scale, and does not by itself establish what happens when the same geometric quantities are moved within a single, fixed

representation by a specific training intervention – the question this study asks for one such intervention, domain-adversarial disentanglement, and one such geometric summary, condition number, among others.

E.6 Reliability Estimation in Medical Imaging Under Distribution Shift

That predictive uncertainty degrades under dataset shift, and that this degradation is consequential for clinical decision-making, is established broadly by Ovadia et al. (2019) and discussed specifically for clinical deployment by Finlayson et al. (2021). In dermatology, out-of-distribution detection for skin lesion images has been examined with deep isolation-forest scoring (Li et al., 2020) and benchmarked systematically across covariate-shifted, near-OOD, and far-OOD regimes (Gutbrod et al., 2025); deep skin-cancer classifiers have also been shown to remain overconfident on out-of-distribution inputs (Pacheco et al., 2020b). A related and largely orthogonal source of distributional heterogeneity in this domain is skin tone: Daneshjou et al. (2022) curated a diverse, pathologically confirmed clinical image set and showed that dermatology AI models trained on conventional benchmarks perform substantially worse on darker skin tones and uncommon diseases, a gap this study’s own dataset pair does not disentangle from acquisition-device shift (Section 5.4). The field has also begun moving from task-specific convolutional classifiers toward large pretrained encoders used as frozen feature extractors in dermatology specifically (Yan et al., 2025); whether the geometry-reliability dissociation reported here persists in such representations is a question this study’s design does not address but that motivates direct follow-up. That literature evaluates reliability estimators against naturally occurring distribution shift; it does not evaluate what happens to those estimators when the representation they operate on is itself the target of an explicit debiasing intervention, which is the setting audited here. This study also does not evaluate calibration (Guo et al., 2017) or selective prediction (El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017), nor distribution-free alternatives to both such as conformal prediction (Angelopoulos & Bates, 2023), directly – these are natural extensions of the reliability question addressed here and are noted as such in Section 5.4.

E.7 Relation to literature, continued

Section 5.3 returns to the two literatures whose relationship to these findings is most direct. This subsection covers the remaining ones and the practitioner implications.

The distinction this study draws between the presence of information in a representation and a specific estimator’s ability to use it is closely related to, and constrained by, the probing-classifier literature’s own caution that a supervised probe’s success does not establish that a model itself uses the decoded information in the way the probe accesses it (Hewitt & Liang, 2019; Belinkov, 2022). This study makes a narrower claim than that literature warns against overreaching toward: the probes here are not offered as evidence about what the disentanglement objective computes internally, only as an independent measurement of what remains extractable from its output representation, without characterizing the mechanism by which that representation was produced.

This finding also bears on the geometry-performance association reported across frozen foundation-model backbones by Janiak et al. (2026) (Section E.5). That association is measured between representations whose provenance – pretraining corpus, objective, architecture, and scale – differs simultaneously, so it cannot by itself distinguish a geometric explanation from a shared dependence on pretraining. The present study varies orthogonality strength within a fixed architecture and representation dimensionality and finds no corresponding geometry–AUROC association for the measured quantities tested here, including condition number (Section 4.2). Whether this absence of association reflects a property specific to domain-adversarial disentanglement, to this dataset pair, or more generally to interventions that hold pretraining fixed, is a question this study’s design cannot resolve on its own and that we regard as a natural target for direct follow-up.

More broadly, these findings suggest that evaluating learned representations only through downstream predictive performance may overlook how a downstream reliability score orders shifted inputs. Raw geometric summaries, scorer-relevant geometry, operational score direction, orientation-free scalar separability, and supervised decodability are related but distinct properties; change in one should not automatically be interpreted as change in the others.

For practitioners combining domain-adversarial representation learning with a distance-based reliability gate, this audit implies a specific caution: a score can weakly distinguish the targeted domain yet assign that distinction the wrong operational polarity. Validation should therefore test both discrimination and direction on calibration data independent of final evaluation, rather than infer operational validity from an orientation-free metric or from general model qualification.

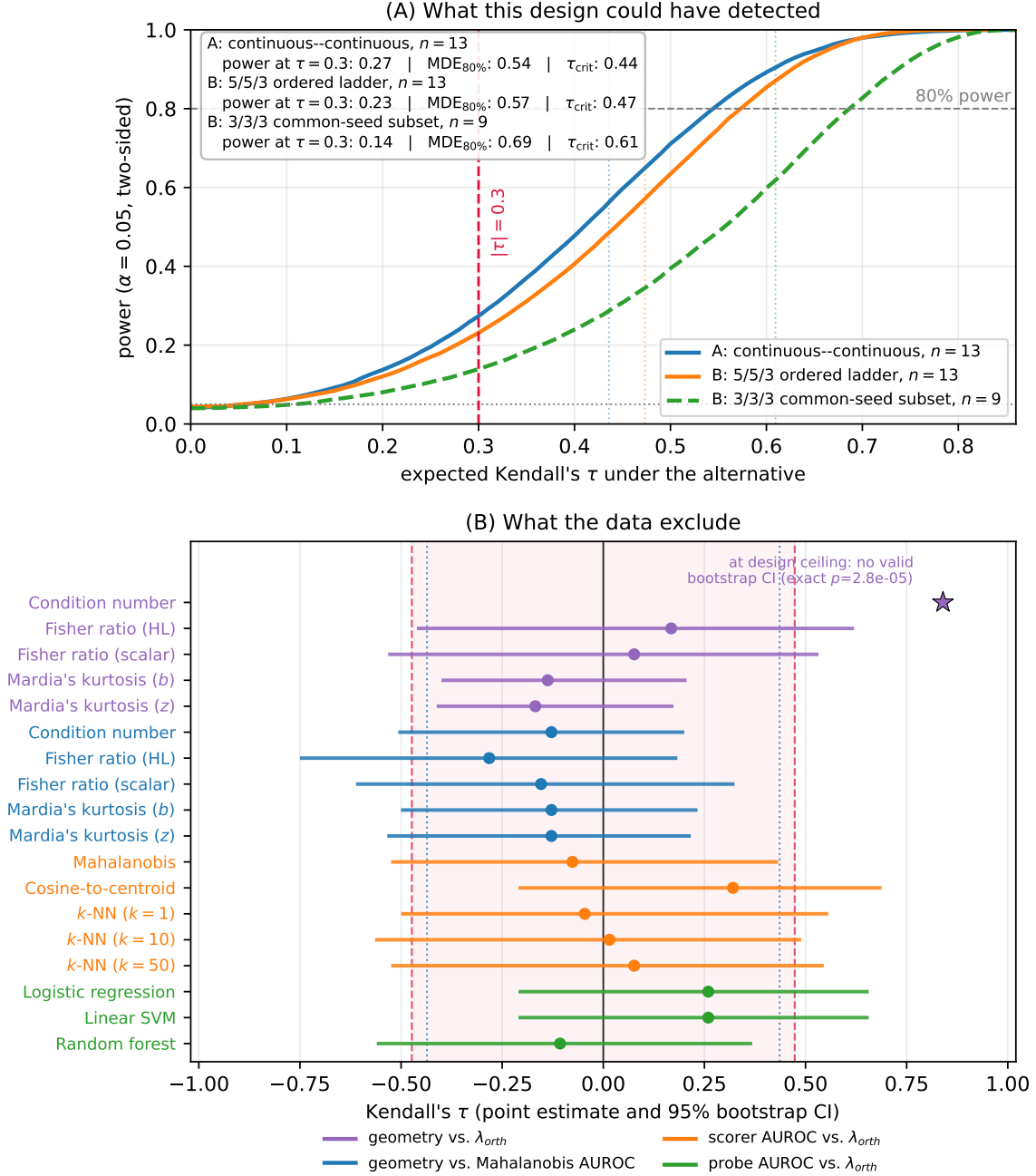


Figure 7: Effect-size and power bounds on this study’s non-significant associations (exact permutation tests, $\alpha = 0.05$). (A) Power against the expected value of the reported statistic, for the three designs of Table 6; vertical dotted lines mark each design’s τ_{crit} and the dashed red line marks $|\tau| = 0.3$, the effect size used in the analysis criterion (Section 3.5). That value lies below every τ_{crit} , so the criterion’s conjunction of $|\tau| \geq 0.3$ with significance at $\alpha = 0.05$ was unsatisfiable at these designs. (B) Observed Kendall’s τ with 95% BCa bootstrap intervals for all eighteen association tests reported in the paper. The shaded band spans $\pm \tau_{crit}$ for the 5/5/3 ladder (dashed bounds); dotted bounds mark the continuous–continuous design. Condition number against λ_{orth} (star) sits at its design ceiling and is drawn without an interval, for the reason given in Section C.2.